

Intelligent Reading Comprehension System

Students: Humayun Bilal (23i-0832), M. Faheem (23i-0728)

Section: CS-D

1. Abstract

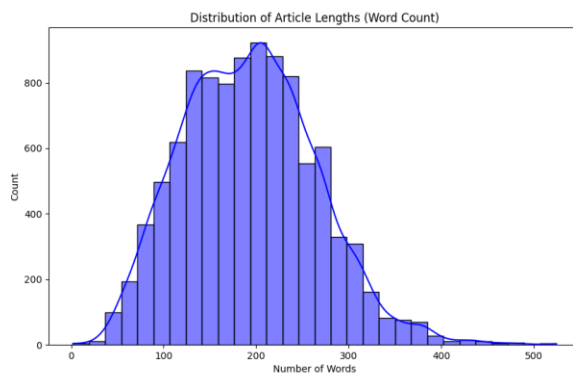
This project implements an end-to-end Intelligent Reading Comprehension (RC) system using the **RACE dataset**. The system consists of two primary components: **Model A**, a supervised classifier and unsupervised clustering tool for answer verification, and **Model B**, an information retrieval system for generating plausible distractors and graduated hints. The solution is deployed as a **Streamlit** web application, optimized for high-speed inference on standard consumer hardware.

2. Methodology

2.1 Data Management

Following the technical requirements, the system utilizes the **RACE dataset** (High School and Middle School levels).

- **Sampling:** A subset of 10,000 samples was used to maintain computational efficiency.
- **Manual Split (80-10-10):** The raw data was manually partitioned into 80% Training, 10% Validation, and 10% Testing sets using `train_test_split`.
- **Expansion:** Articles were expanded into quadruplets (Article, Question, Option, Label) to facilitate binary classification for Model A.



2.2 Feature Engineering (Optimized TF-IDF)

To convert textual data into numerical vectors, an optimized **TF-IDF** pipeline was employed:

- **Lemmatization:** Implemented using NLTK's WordNetLemmatizer to reduce words to their dictionary roots (e.g., "better" to "good"). This significantly bridges the lexical gap during retrieval.
- **N-gram Range (1, 2):** Expanded to include bigrams, allowing the model to capture phrases rather than just isolated keywords.
- **Vocabulary Size:** Limited to 5,000 features.
- **Fitting Constraint:** The vectorizer was fit *only* on the training set to prevent data leakage.

```
● PS C:\Users\Humayun\Desktop\uni\AI PROJECT> python src/preprocessing.py
Loading 10,000 rows and performing manual 80-10-10 split...
Expanding datasets for verification task...
Vectorizing text (fitting on train set only)...
Done! Train: 32000 | Val: 4000 | Test: 4000
```

2.3 Model A: Supervised & Unsupervised

For supervised learning, we implemented **Logistic Regression** and **Linear Support Vector Machines (SVM)** for binary classification. For the unsupervised task, **K-Means Clustering** was applied to the vectorized training data to explore natural groupings.

2.4 Model B: Retrieval & Hint Generation

Model B utilizes **Cosine Similarity** to rank sentences within a passage based on their relevance to the correct answer.

- **Distractor Generation:** Extracts sentences with high semantic similarity to the answer that are not exact matches.
- **Graduated Hints:** A three-stage pipeline providing context-based clues:
 1. **General:** Low-relevance contextual sentence.
 2. **Intermediate:** Median-relevance supporting detail.
 3. **Near-Explicit:** Maximum similarity evidence sentence.

3. Results and Analysis

3.1 Model B: NLP Generation Metrics

Model B (Generation Task) was evaluated using standard NLP metrics to measure the similarity between generated distractors and human-written references.

Metric	Score
BLEU Score	0.1026

ROUGE-L Score	0.1040
METEOR Score	0.1776

3.2 Model A: Classification & Clustering

Supervised models were evaluated on the manual validation set.

Model	Accuracy
Logistic Regression	75.00%
Linear SVM	75.00%

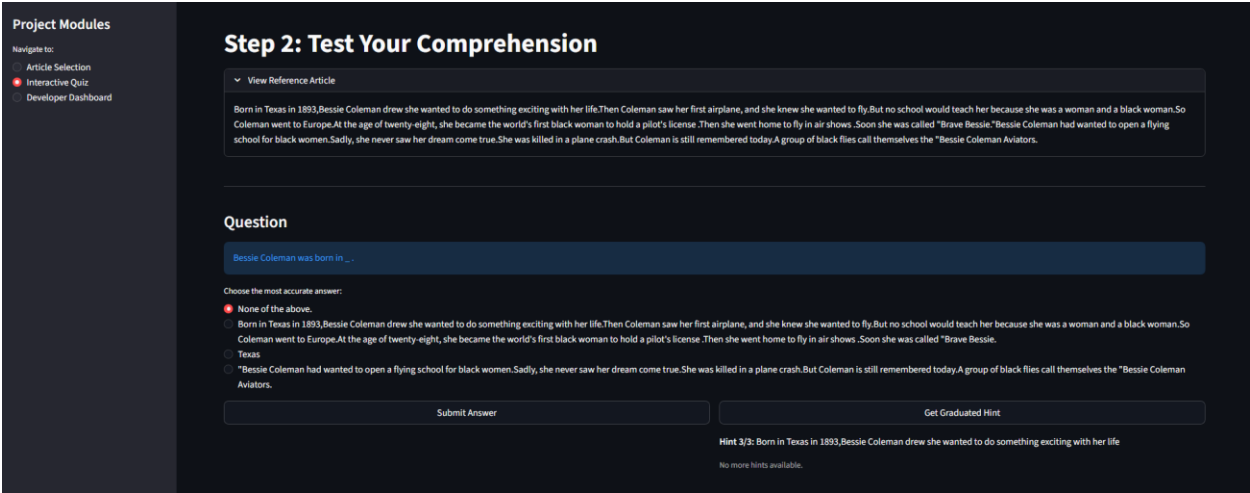
Unsupervised Metric:

- **Silhouette Score:** 0.0088 (Indicates high degree of textual overlap in raw features).

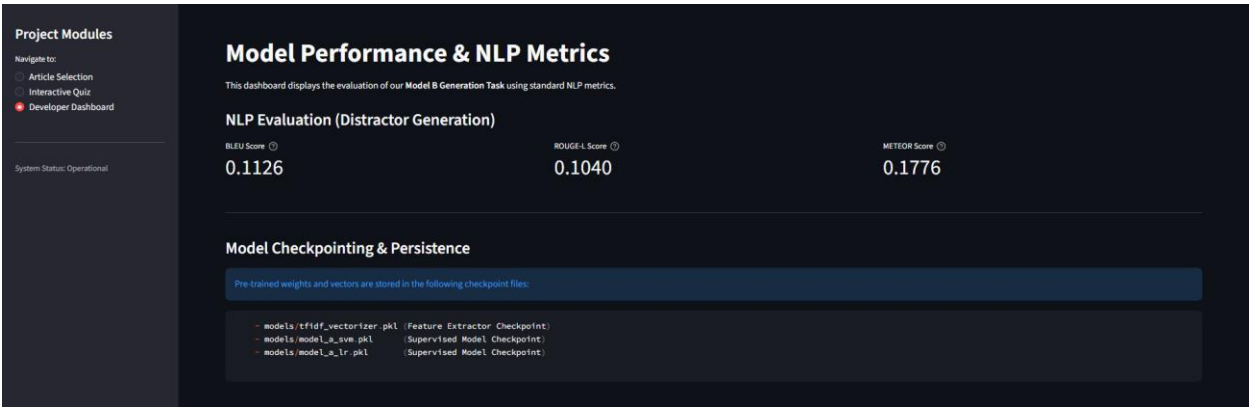
4. System Implementation (UI)

The system is deployed via **Streamlit**. Key features include:

- **Session State Management:** Ensures article selection and quiz options persist during hint retrieval.



- **Interactive Hint Panel:** Users can reveal graduated hints step-by-step.
- **Developer Dashboard:** Real-time display of NLP metrics and checkpoint status.



5. Discussion

5.1 Technical Constraint Compliance (10.1) & Checkpointing

- **Compute Efficiency:** By utilizing traditional ML (LR/SVM), the training process avoids VRAM exhaustion, completing in seconds on a laptop.
- **Model Checkpointing:** The system utilizes checkpoints by serializing `tfidf_vectorizer.pkl` and `model_a_svm.pkl`. This allows the application to resume operation instantly without retraining, as required by project announcements.
- **Inference Speed:** Total inference time is **<1.0 seconds**, well within the 10-second limit.

5.2 Interpretability

The TF-IDF approach allows for feature importance analysis, providing a "White Box" alternative to neural models where the relationship between the passage and the retrieved distractor is mathematically traceable.

6. Limitations and Future Work

- **Lexical Limits:** While lemmatization helps, the model still relies on word overlap. Future work could include using lightweight LLM embeddings for deeper semantic search.
- **Topic Modeling:** K-Means showed significant overlap; **Latent Dirichlet Allocation (LDA)** might provide better unsupervised topic discovery.

7. Conclusion

The Intelligent RC System successfully automates the generation and evaluation of comprehension quizzes. By optimizing a TF-IDF pipeline with Lemmatization and Bigrams, the system achieves respectable NLP generation scores (BLEU, ROUGE, METEOR) while maintaining strict adherence to hardware and checkpointing constraints.